



Database Project Ideation

IT214 – Database Management Systems

Textile Manufacturing Operations Management System

Faculty Instructor

PM Jat

Project Team

No.	Name	Student ID
1	Sujal Balva	202401216
2	Yash Agarwal	202401247
3	Raj Kachhadiya	202401255
4	Madhav Parmar	202401260
5	Parthiv Panchal	202401259

Group Representative

Sujal Balva (202401216)

Contact: +91 8160725965

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Objective

The objective of this project is to design a database system for a textile manufacturing mill that produces fabric materials and supplies them in bulk to large-scale garment manufacturing companies. The mill operates through several departments responsible for procurement of raw materials, production of fabric, quality inspection, packing, warehouse management, sales, and logistics.

The database system will maintain information about employees, departmental activities, raw materials, production processes, inventory levels, bulk orders from garment manufacturers, and shipment of finished fabric products. The system will help different departments coordinate their operations efficiently and enable managers to monitor production, workforce allocation, inventory availability, and order fulfillment.

Application Users

The proposed database system will be used by several stakeholders within the textile manufacturing organization.

1. **Company Administrator** – Maintains overall system information and supervises operations across departments.
2. **HR Managers** – Manage employee records, department assignments, attendance, and workforce information.
3. **Production Managers** – Monitor production activities, machines used, and quantities of fabric produced.
4. **Quality Control Inspectors** – Record inspection results, identify defects, and approve or reject fabric batches.
5. **Warehouse Managers** – Track storage of finished fabric rolls and maintain warehouse inventory.

6. **Sales Managers** – Manage interactions with garment manufacturers that place bulk orders.
7. **Logistics Managers** – Handle shipment and delivery of finished fabric products.

Organizational Structure

The textile mill operates through multiple departments that coordinate together to manage the manufacturing and distribution of fabric. The database system maintains information about these departments and the employees assigned to them.

Typical departments in the textile mill include:

- Procurement Department
- Production Department
- Dyeing and Processing Department
- Quality Control Department
- Packing Department
- Warehouse Department
- Logistics Department
- Sales Department
- Maintenance Department
- Human Resources Department

Each department contains employees with specific roles and responsibilities, and managers supervise departmental operations.

Use Cases – HR Department

The HR department manages employee-related information within the organization.

- Maintain employee records including personal and professional details.
- Assign employees to specific departments and roles.
- Track employee shifts and work schedules.
- Maintain attendance and employment history of workers.
- Monitor workforce distribution across departments.

Use Cases – Procurement Department

The procurement department ensures that sufficient raw materials are available for textile production.

- Maintain information about suppliers providing raw materials.
- Record purchase orders placed for materials such as cotton, yarn, synthetic fibers, dyes, and chemicals.
- Track delivery of raw materials from suppliers.
- Maintain records of raw material inventory levels.
- Monitor material usage across production activities.

Use Cases – Production Department

The production department converts raw materials into finished fabric through multiple manufacturing stages.

- Maintain information about machines used in production.
- Assign machine operators to production activities.

- Record production batches created during manufacturing.
- Track the type of fabric produced in each batch.
- Monitor the quantity of fabric produced over time.

Typical production processes include spinning, weaving or knitting, dyeing, and finishing.

Use Cases – Quality Control Department

The quality control department ensures that produced fabric meets the required quality standards before being shipped.

- Record inspection of fabric batches.
- Maintain information about inspectors performing quality checks.
- Record defects identified during inspections.
- Approve or reject fabric batches based on quality standards.
- Maintain historical records of inspection results.

Use Cases – Packing and Warehouse Management

After quality approval, finished fabric is packed and stored before shipment.

- Record packing of fabric rolls and quantities packed.
- Maintain warehouse inventory of finished fabric products.
- Track storage locations of fabric rolls.
- Monitor stock availability in the warehouse.

Use Cases – Sales Department

The textile mill sells its fabric in bulk to garment manufacturing companies.

- Maintain information about garment manufacturing companies purchasing fabric.
- Record bulk orders placed by manufacturers.
- Track order details including fabric types and quantities requested.
- Maintain historical records of orders placed by manufacturers.

Use Cases – Logistics and Shipment

The logistics department manages transportation of finished fabric to manufacturing companies.

- Record shipment details for dispatched orders.
- Maintain information about transport companies used for delivery.
- Track shipment status and delivery confirmation.
- Maintain shipment history for bulk orders.

Tentative List of Reports (Queries)

The database system should support queries that allow managers to monitor operations and analyze performance.

1. Employees working in each department.
2. Production statistics for different fabric types.
3. Raw material inventory levels.
4. Fabric batches rejected due to quality defects.

5. Current stock of fabric rolls in warehouses.
6. Bulk orders placed by garment manufacturing companies.
7. Shipment history and delivery status of orders.

These reports help the organization monitor production efficiency, workforce allocation, inventory availability, and order fulfillment.